CRISISAID ROUTE PLANNER

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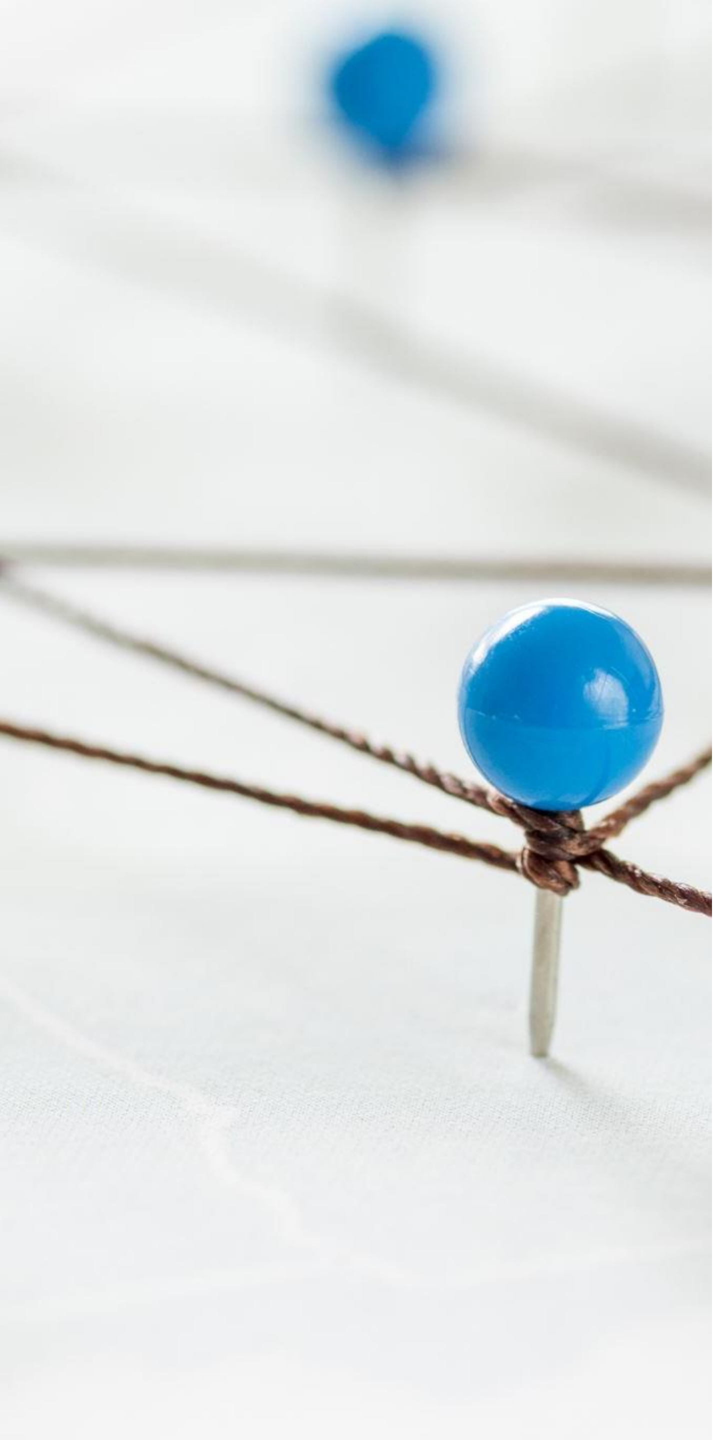
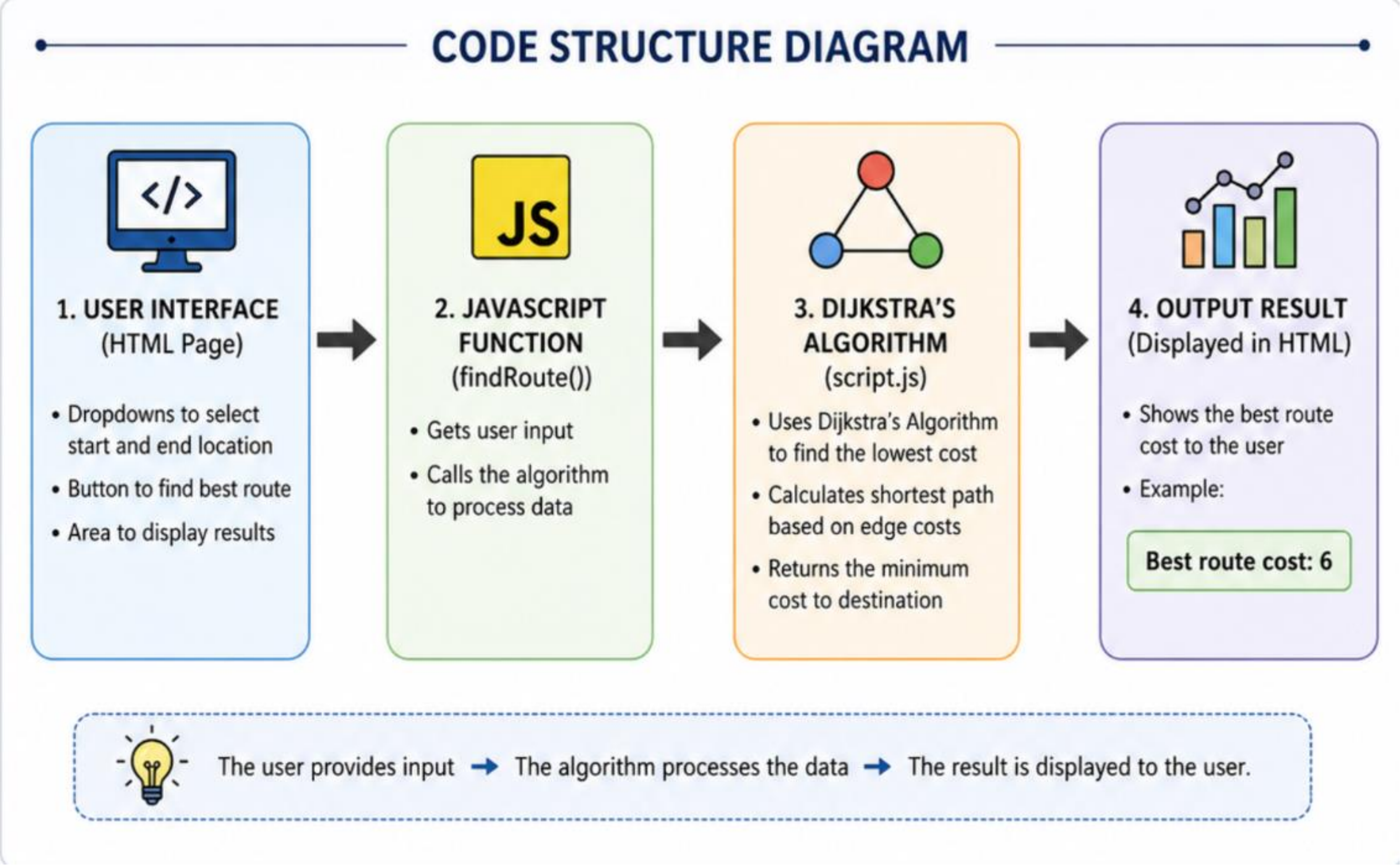


PROJECT OVERVIEW

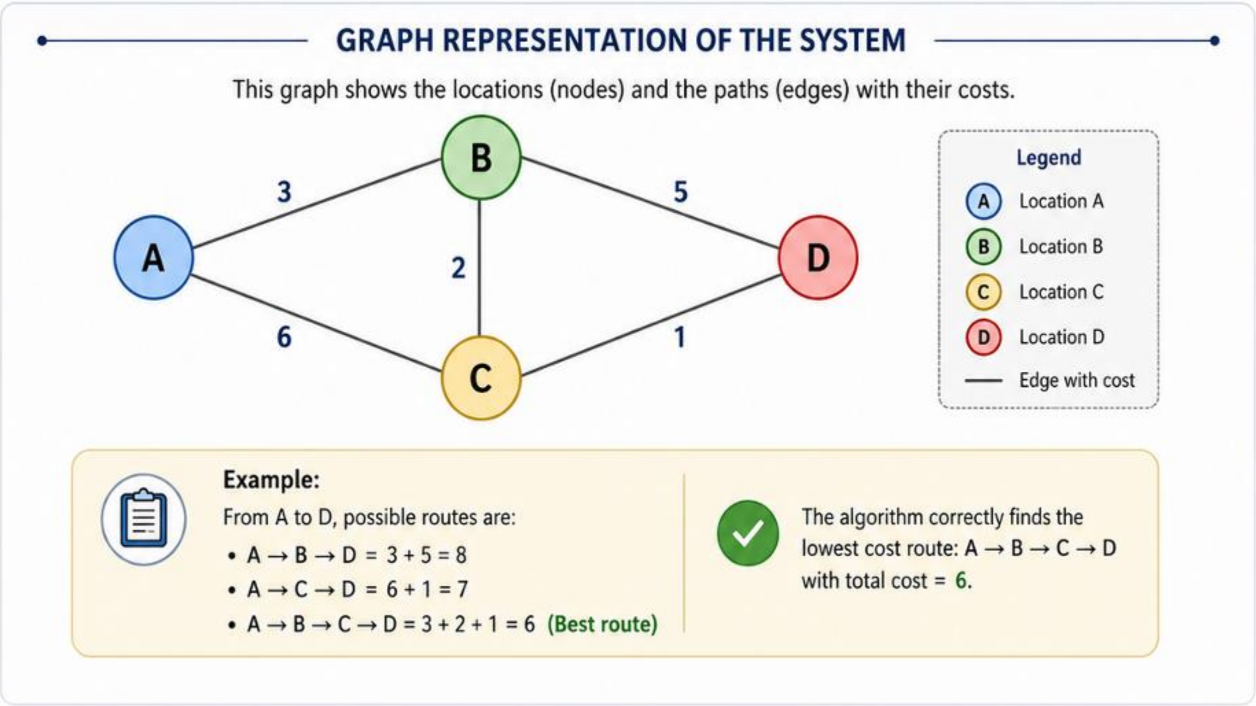
- Web-based application for route planning
- Designed for emergency relief scenarios
- Finds the most efficient path between locations
- Uses a graph-based model with weighted edges
- Focus: distance, risk, and travel cost optimization

SIGNIFICANCE OF THE PROJECT

- Helps simulate real-world disaster decision-making
- Supports safe and efficient supply delivery planning
- Uses simplified nodes (A, B, C, D) to represent locations
- Edge weights represent: Distance, risk level, travel difficulty
- Shows how algorithms improve decision-making


CODE STRUCTURE DIAGRAM
1. USER INTERFACE
(HTML Page)
• Dropdowns to select start and end location
• Button to find best route
• Area to display results
JS
2. JAVASCRIPT FUNCTION
(findRoute())
• Gets user input
• Calls the algorithm to process data
3. DIJKSTRA'S ALGORITHM
(script.js)
• Uses Dijkstra's Algorithm to find the lowest cost
• Calculates shortest path based on edge costs
• Returns the minimum cost to destination
4. OUTPUT RESULT
(Displayed in HTML)
• Shows the best route cost to the user
• Example:
Best route cost: 6
The user provides input → The algorithm processes the data → The result is displayed to the user.

ALGORITHM USED



- **Dijkstra's Algorithm**
- Finds the shortest path in a weighted graph
- Steps:
  - Start from the initial node
  - Assign initial cost = 0
  - Visit the smallest cost node
  - Update neighboring nodes
  - Repeat until the destination is reached
- Output: minimum-cost route

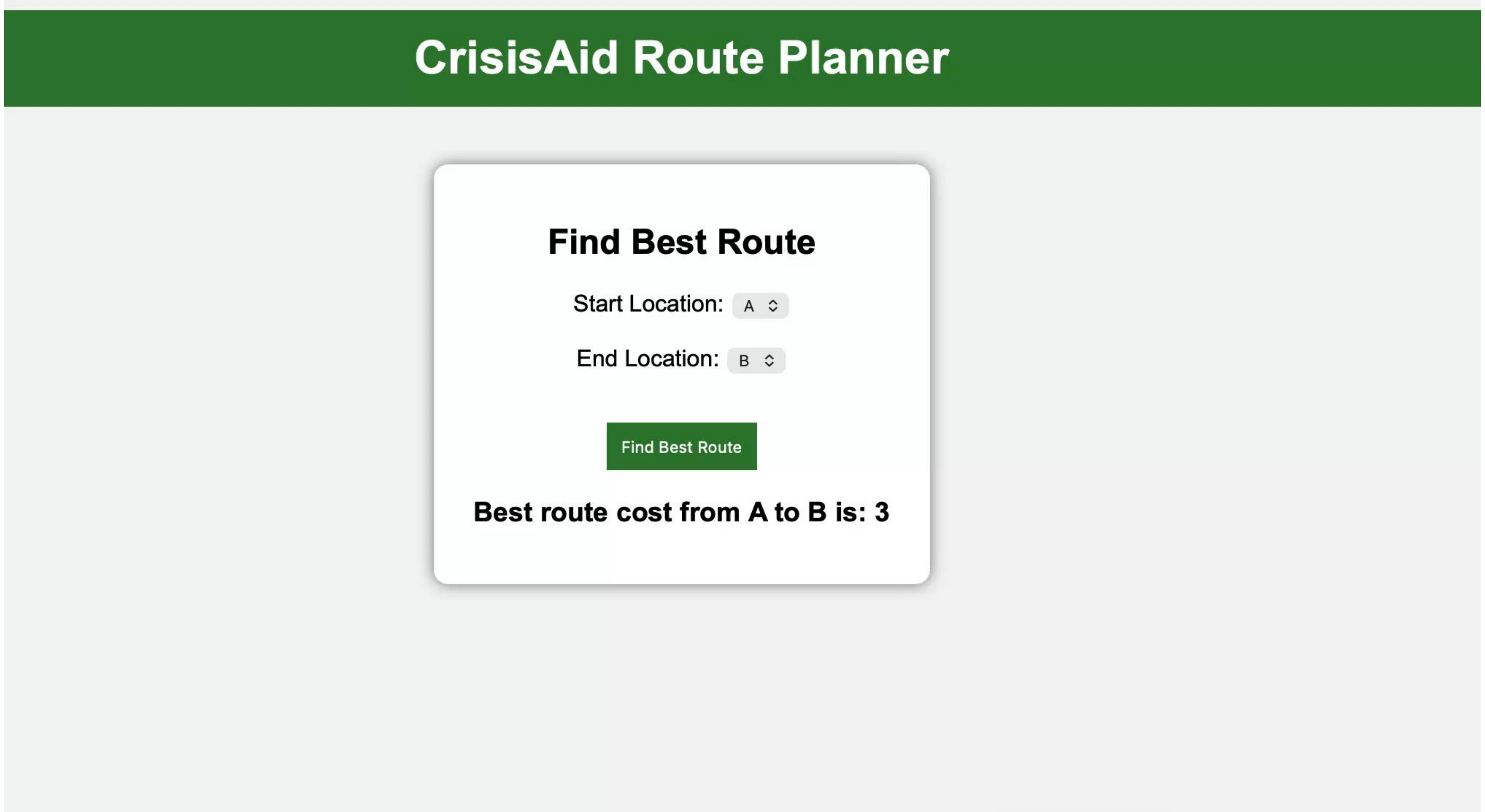
CONCLUSION

The project demonstrates route optimization using Dijkstra's Algorithm.

It finds the most efficient path between locations.

It shows how algorithms can support better decision-making in emergencies.

VIDEO


CrisisAid Route Planner
Find Best Route
Start Location: A
End Location: B
Find Best Route
Best route cost from A to B is: 3